III. SOFTWARE ARCHITECTURE DESIGN

3.1. Architecture Selection

The Phone Book Management System will use a **3-Tier Architecture** combined with the **Model-View-Controller (MVC) design pattern**.

This architecture separates the system into three main layers: Presentation Layer, Business Logic Layer, and Data Access Layer. Each layer has its own responsibilities and communicates with the other layers to perform system operations.

The MVC pattern is applied within the Presentation Layer to separate the user interface, application logic, and data management.

This architecture is suitable for the Phone Book Management System because it improves code organization, simplifies maintenance, enhances security, and supports future system expansion.

3.2. Architecture Components

1. Presentation Layer (User Interface)

This layer provides the graphical user interface through which users interact with the system.

Main responsibilities:

- Display login and registration pages.
- Display contact lists and contact details.
- Provide interfaces for adding, editing, and deleting contacts.
- Provide interfaces for managing Tags and Favourite Lists.
- Provide CSV contact import functionality.
- Provide administrative interfaces for managing user accounts and viewing system statistics.
- Display success messages, error messages, and operation results.

2. Business Logic Layer (Application Logic)

This layer processes user requests and implements the main business rules of the system.

Main responsibilities:

- Authenticate users and manage login sessions.
- Verify user permissions based on User and Admin roles.
- Validate contact, tag, and account information.
- Process adding, editing, searching, and deleting contacts.
- Manage Tags and Favourite Lists.
- Process CSV files and validate imported contact records.
- Handle user account management operations.
- Generate system statistics.
- Ensure that users can only access and modify their own data.

3. Data Access Layer (Database Management)

This layer is responsible for communicating with the database and managing data storage and retrieval.

Main responsibilities:

- Store and retrieve user account information.
- Store and manage contact information.
- Store Tags and Favourite List information.
- Maintain relationships between Users, Contacts, Tags, and Favourite Lists.
- Execute database queries for searching, updating, and deleting data.
- Ensure data consistency and referential integrity.

3.3. MVC Design Pattern

Within the Presentation Layer, the system applies the MVC design pattern, which consists of three components:

- **Model:** Represents system data and manages interactions with the database, including User, Contact, Tag, and Favourite List.
- **View:** Displays the user interface, including login pages, contact lists, contact forms, and administrative dashboards.
- **Controller:** Receives user requests, processes actions, communicates with the Business Logic Layer, and returns results to the View.

3.4. System Architecture Workflow

The system follows this general processing flow:

1. The user interacts with the system through the Presentation Layer.
2. The Controller receives the user's request.
3. The Business Logic Layer processes the request and validates the necessary information.
4. The Data Access Layer communicates with the database to retrieve or update data.
5. The database returns the requested results.
6. The results are passed back through the Business Logic Layer and Controller.
7. The View displays the results to the user.

3.5. Advantages of the Selected Architecture

- **Modularity:** Separating the system into layers makes the source code more organized.
- **Maintainability:** Developers can modify one layer without significantly affecting other layers.
- **Security:** Authentication and authorization can be handled in the Business Logic Layer.
- **Scalability:** New features can be added without major changes to the entire system.
- **Reusability:** Business logic can be reused by different functions of the system.

- **Flexibility:** The system can be modified or expanded to support future requirements.

3.6. System Architecture Diagram

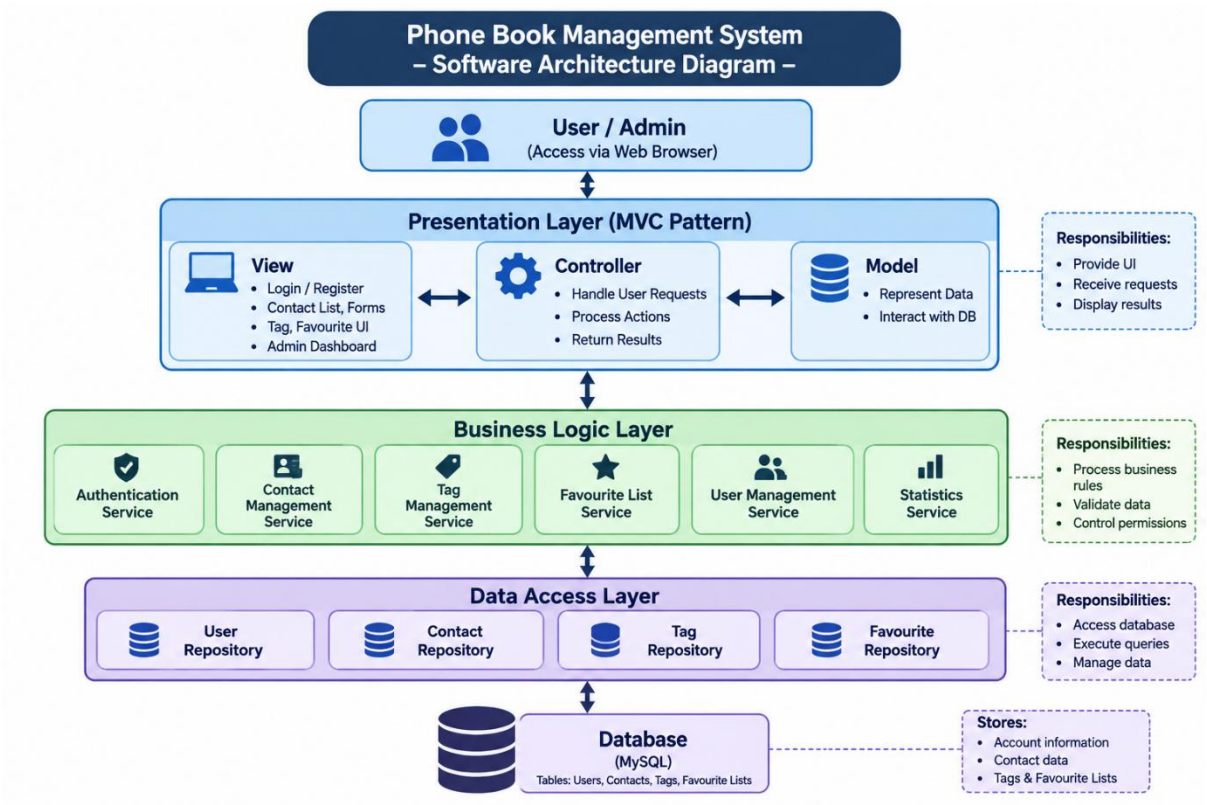


Figure 3.6. Software Architecture Diagram of the Phone Book Management System

The diagram illustrates the three-tier architecture of the system, consisting of the Presentation Layer, Business Logic Layer, and Data Access Layer. It also demonstrates how users and administrators interact with the system and how data is processed and stored in the database.